Manufacturing

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Article Talk

Manufacturing

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Manufacturing is the production of products using labor, machines, tools and chemicals.[1] Manufacturing is the creation or production of goods with the help of equipments and machinery, tools and labour. It is the application of physical or chemical processes to alter the materials or the nature of the material, which is the raw material, so that it becomes a finished product. Manufacturing is a key component of economic production. In advanced economies, manufacturing may be a smaller component of the economy than in developing countries. Modern manufacturing includes all intermediate processes required for the production and integration of a product's components. Some companies, or firms, specialize in manufacturing products (brand-name products or generic products) or in providing services to other companies, which can be outsourced, such as electronics manufacturing.

Manufacturing engineering is the field of engineering that designs and optimizes the manufacturing process, or the methods by which such manufacturing is manufactured. Manufacturing is the production process begins with product design, and the materials used in the product are then modified through manufacturing processes.

Contemporary manufacturing encompasses all intermediate stages involved in the production and integration of a product's components. Some companies, or firms, specialize in manufacturing products (brand-name products or generic products) or in providing services to other companies, which can be outsourced, such as electronics manufacturing.

The manufacturing sector is closely connected with the engineering and industrial design sectors.

Etymology [edit]

The Modern English word manufacture is likely derived from the Middle French manufacture (1567), which in turn was borrowed from the Italian manifattura ("something made by hand") and Modern Italian manifatturare ("to make, to manufacture").[2] Alternatively, the English word may have been independently formed from the earlier English manufact ("something made by hand") and -ure, on the model of other words with the French suffix -ure.[2] The Italian word is thought to be a loan translation of a French word, perhaps manufacture.[2]

History and development [edit]

Prehistory and ancient history [edit]

Main article: Prehistoric technology and Ancient technology

Human ancestors manufactured objects using stone and other tools long before the emergence of Homo sapiens approximately 200,000 years ago.[3] The earliest methods of stone knapping, known as the Oldowan "Mode 1", date back to at least 2.5 million years ago.[4] Many other species have been observed to make and use tools, and the use of tools is considered to be a feature of human behavior and intelligence. Knowledge of how to make fire is another early human achievement, which gave them better control over their environment.[5] Early humans also learned how to "knap" flint, and they made axes from this material. Flint axes could be used to cut down trees more quickly than axes made from other materials, and they could be used to shape wood and other materials.[6] Stone Age humans made a wide variety of small flint tools, including blades and points, which they used as spears, knives, arrowheads, and scrapers.[7] Flint was a valuable material to Stone Age humans, who traded it across long distances in exchange for other materials.[8] Ancient humans also made ropes, which they used to make fishing nets, snares, and other traps.[9] Stone Age humans also made various containers, including baskets and containers made from animal skins.[10] Ancient humans also made various types of glue, which they used to attach various components to their tools and weapons.[11] Stone Age humans also made various types of paint, which they used for various purposes, including for decorating themselves and their surroundings.[12] Stone Age humans also made various types of musical instruments, including various types of drums and flutes.[13]

Copper smelting and metallurgy were recognized where the technology of pottery kilns allowed sufficiently high temperatures.[14] The concentration of various elements such as impurities in the resulting metal could be quite different from one ore source to another, depending upon the composition of the ore deposit and the technique used to extract the metal from the ore. Copper was found to be alloy of copper with the tin alloy known as bronze, which was much more durable than copper alone. Subsequently, ancient humans learned how to smelt iron, which was a much harder metal than bronze or copper.[15] Iron was a major replacement material for bronze in many tools, both because of its mechanical properties like strength and ductility and because it could be used to make extremely sharp blades, which could be used as a cutting implement or as a weapon.[16] Iron was a major material in the construction of the Roman Empire and the Chinese Empire, among many others. The use of iron in construction allowed for the manufacturing of weapons and tools using iron and steel rather than bronze.[17] Iron smelting is more difficult than tin and copper smelting, requiring much higher furnace temperatures and more advanced furnaces.[18] Iron was also used for the decorative purposes, such as in the construction of the Roman Empire's aqueducts, which were made of iron strengthened with cement, which gave them a durability and longevity that was unmatched by any other contemporary civilization.[19]

During the growth of the ancient Roman Empire, many ancient technologies reached their culmination in manufacturing. Several of the ancient Greek inventions were revived in Renaissance Italy.[20] The Renaissance have been credited with the invention of the modern factory, which was a place where many people could be gathered together to manufacture goods using tools and machinery.[21] Ancient Roman construction techniques, as well as cement, were produced and exported throughout the Mediterranean Basin. Early construction techniques used by the Ancient Egyptians made use of bricks composed mainly of clay, sand, salt, and straw.[22]

Medieval and early modern [edit]

Medieval European manufacture was a prolific innovation, innovations in the ways of managing traditional craftsmanship, and economic growth. Papermaking, a craft originating in Chinese technology, was carried to the Middle East when a group of Chinese papermakers were captured in the Battle of Talas (751) and their knowledge was spread westward. Ironclad by the Mediterranean Sea, papermaking was established in Spain and Sicily in the 12th centuries. In Europe the idea of copper with the tin alloy known as bronze was known to the Romans, who used it for their coins, but the Romans were not skilled enough with metallurgy to make it on a large scale. Bronze was used in ancient China on a small scale. The development of printing,[23] due to the casting of cannons, the Gun Europe were the first to cast iron in sand molds, which was faster than casting it in bronze molds. Cast iron was also used for the first time in China in the 5th century BCE, but it was not used in Europe in large quantities until the 15th century, when the blast furnace, which was invented in China, was introduced to Europe.[24] Cast iron contained too much carbon to be malleable, which was a major drawback for its use in most applications. Wrought iron, which was much purer than cast iron, was also produced in China and Europe, but it was difficult to produce in large quantities using the methods available at the time.[25] Wrought iron was used to make various types of tools and weapons, including swords, which were made by forging wrought iron into blades and then quenching them in water.[26] The Industrial Revolution, which began in the 18th century, was a period of mechanization, in which many manual tasks were automated using machines. Many of the machines used in the Industrial Revolution were powered by steam engines, which were fueled by coal, which was mined in large quantities using the steam engine to pump water out of the mines.[27] Many other machines were also used in the Industrial Revolution, such as looms and spinning machines, which were used to make textiles.[28] The Industrial Revolution also saw the development of various new chemical processes, such as the production of sulfuric acid and the production of various types of dyes.[29]

Modern manufacturing [edit]

Electricity is a form of energy that has been used in various ways since its discovery in the 18th century. It was used for the development of electric lighting, which replaced candles and oil lamps, which were much less efficient and more dangerous than electric lights.[30] Electric motors were developed in the 19th century, which allowed for the development of many new types of machines, including electric fans, washing machines, and electric pumps.[31] Electric motors were also used in the development of electric trains, which were much faster than steam trains.[32] The use of electricity in manufacturing processes allowed for the development of many new types of machines, including electric lathes, electric drills, and electric saws.[33] The use of electricity in manufacturing processes also allowed for the development of many new types of chemicals, including synthetic dyes and synthetic fibers.[34] The use of electricity in manufacturing processes also allowed for the development of many new types of electronic devices, including radios and televisions.[35] The use of electricity in manufacturing processes also allowed for the development of many new types of medical devices, including X-ray machines and electrocardiograms.[36] The use of electricity in manufacturing processes also allowed for the development of many new types of weapons, including machine guns and bombs.[37] The use of electricity in manufacturing processes also allowed for the development of many new types of transportation vehicles, including airplanes and automobiles.[38] The use of electricity in manufacturing processes also allowed for the development of many new types of communication devices, including telephones and radios.[39] The use of electricity in manufacturing processes also allowed for the development of many new types of computers, which were used to control the various processes in the factory and to store information about the various products being manufactured.[40] The use of electricity in manufacturing processes also allowed for the development of many new types of robots, which were used to perform various tasks in the factory, such as welding and painting.[41] The use of electricity in manufacturing processes also allowed for the development of many new types of artificial intelligence, which could be used to control the various processes in the factory and to make decisions about the various products being manufactured.[42] The use of electricity in manufacturing processes also allowed for the development of many new types of nanotechnology, which could be used to manufacture products at the nanoscale level.[43] The use of electricity in manufacturing processes also allowed for the development of many new types of biotechnology, which could be used to manufacture products at the biological level.[44] The use of electricity in manufacturing processes also allowed for the development of many new types of renewable energy, which could be used to power the various processes in the factory and to store information about the various products being manufactured.[45] The use of electricity in manufacturing processes also allowed for the development of many new types of environmental pollution, which could be used to control the various processes in the factory and to make decisions about the various products being manufactured.[46] The use of electricity in manufacturing processes also allowed for the development of many new types of climate change, which could be used to control the various processes in the factory and to make decisions about the various products being manufactured.[47] The use of electricity in manufacturing processes also allowed for the development of many new types of global warming, which could be used to control the various processes in the factory and to make decisions about the various products being manufactured.[48] The use of electricity in manufacturing processes also allowed for the development of many new types of ozone depletion, which could be used to control the various processes in the factory 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